Mahdi Farsi

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Professional Summary

PhD in Mechanical Engineering with expertise in computational fluid dynamics (CFD), turbulence modeling, experimental fluid mechanics, heat transfer, and thermal management of electronic systems. Experienced in numerical simulation, wind tunnel experimentation, and custom data analysis using Python, MATLAB, and Fortran.

Professional Experience

Postdoctoral Fellow, University of Houston, Houston, TX — Mar. 2026 – Present

- Conducting wind tunnel experiments on the wake of vertical-axis wind turbines using *Particle Image Velocimetry (PIV)* and high-speed imaging systems.
- Characterizing active-grid turbulence through automated traverse motion, hot-wire measurements, and Python-based data acquisition with NI-DAQ and Arduino control.
- Designing and 3D-printing scaled vertical-axis wind turbine blades for experimental wake measurements.

R&D Intern, Valmet, Houston, TX — Jul. 2025 – Feb. 2026

- Performed CFD simulations to evaluate and improve thermal management of the MAXUM II gas chromatograph electronics enclosure using *Ansys Fluent* and *Icepak*.
- Conducted laboratory experiments to validate CFD predictions and optimized fan placement, enhancing internal airflow circulation and reducing peak temperatures on critical electronic surfaces by up to **15 °C**.

Graduate Research Assistant, University of Houston, Houston, TX — Jan. 2021 – Dec. 2025

- Developed *custom LES codes* for pollutant dispersion in atmospheric and oceanic environments.
- Designed and validated cognitive search algorithms for autonomous robots addressing the *inverse problem* of pollutant source localization.
- Conducted CFD analysis for complex industrial applications such as *Master Flo Valve choke valves* and a novel design for *MOCVD reactors* using *Ansys Fluent*.
- Performed open-channel flow simulations with *two-phase modeling* in *Ansys Fluent* to assess repurposing retired wind turbine blades as bridge piers.

Graduate Teaching Assistant, University of Houston, Houston, TX — Jan. 2021 – Apr. 2025

- Assisted instruction for Fluid Dynamics, Heat Transfer, Thermodynamics, and Experimental Methods; mentored students and supported laboratory instruction.

Education

Ph.D., Mechanical Engineering, University of Houston, Houston, TX — Jan. 2021 – Dec. 2025
M.Sc., Mechanical Engineering, University of Tehran, Tehran, Iran — Aug. 2016 – Jan. 2019
B.Sc., Mechanical Engineering, Sharif University of Technology, Tehran, Iran — Aug. 2012 – Jul. 2016

Technical Skills

Programming: Python, MATLAB, Fortran, C
Simulation Software: Ansys Fluent, Icepak, CONVERGE CFD
CAD Tools: SolidWorks, Ansys SpaceClaim
AI/ML Tools: TensorFlow, Keras
Experimental Techniques: Wind tunnel testing, Particle Image Velocimetry (PIV), hot-wire anemometry, sonic anemometry, thermal instrumentation
Instrumentation & Control: NI-DAQ, Arduino, programmable power supplies, traverse automation

Awards & Honors

- Bidani Scholarship (2024), General Scholarship (2023), ABS Academic Excellence (2022), FFP Fellowship (2021)

Publications

- **Farsi, M.**, Yang, D. (2024), Cognitive search algorithms, *APS Bulletin*
- **Farsi, M.**, et al. (2023), LES of bay–channel interactions, *APS DFD Meeting*
- **Farsi, M.**, et al. (2021), Low-speed wind energy harvesting, *Int. J. Env. Sci. Tech.*
- Khoshvaght-Aliabadi, **Farsi, M.**, et al. (2021), Minichannel heat sinks, *Int. Commun. Heat Mass Transf. 122*
- **Farsi, M.**, et al. (2021), Heat transfer in flattened tubes, *Int. J. Therm. Sci. 160*
- Khoshvaght-Aliabadi, **Farsi, M.**, et al. (2021), Modified turbulator heat transfer, *Int. Commun. Heat Mass Transf. 120*
- **Farsi, M.**, et al. (2020), Radial injection in TC–Poiseuille flow, *Numer. Heat Transf. B 77*